Sonoluminescence as Coherence Breakdown: A Non-Markovian Emission Mechanism in Driven Bubble Collapse

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Abstract

The physical mechanism underlying sonoluminescence remains an open problem. Standard descriptions attribute the flash to extreme heating, plasma formation, or thermal bremsstrahlung during rapid bubble collapse. While such models reproduce the gross timing of emission near minimum radius, they do not provide a unified dynamical account of why energy becomes so sharply localized, why the emission is so strongly thresholded, or why timing and intensity can be sensitive to driving conditions.

We present a coherence-response model in which emission arises from the finite-response deformation and breakdown of coherence in a driven medium. A dimensionless coherence variable χ is introduced to parametrize the ability of the medium to sustain organised collective response under forcing. The equilibrium coherence state $\chi_{\rm eq}$ is defined from the competition between intrinsic response and collapse-driving timescales, while the realised coherence χ evolves toward this target through a finite-response relaxation equation. The resulting system is non-Markovian and history-dependent.

Numerical simulations show that emission occurs within a narrow window near collapse, but is controlled by the evolution of χ rather than by instantaneous geometric or thermal extrema alone. In the fast-response regime, emission is effectively coincident with minimum radius, reproducing the appearance of standard collapse-locked timing. However, when the coherence relaxation time is increased, a distinct coherence-lag regime appears in which the emission peak shifts away from minimum radius and follows the delayed coherence evolution instead. This separates the trigger of emission from its governing mechanism.

Degeneracy tests further show that removing thermal contributions does not eliminate emission, while modifying damping and coherence thresholds changes the timing and structure of the flash. These results support a picture in which collapse loads energy into a stressed medium, while coherence deformation and threshold release govern when that energy is radiated.

The model provides a framework for interpreting sonoluminescence as a coherence-mediated energy release process in a structured, finite-response medium. It also yields falsifiable predictions for timing offsets, threshold behaviour, and parameter-dependent phase shifts that can distinguish coherence-driven emission from purely thermodynamic descriptions.

1 Introduction

Sonoluminescence is the phenomenon in which a gas bubble, driven by an acoustic field, emits short bursts of light during collapse. Despite decades of theoretical and experimental study, the detailed mechanism of emission remains debated. Standard explanations include shock-induced heating, thermal bremsstrahlung, plasma formation, and related collapse-induced radiative processes. These models capture important aspects of the gross energetics and near-collapse timing, but they do not yet provide a universally accepted microscopic account of how the mechanical energy of a driven bubble is converted into a sharply localized, rapidly released emission event.

The difficulty is not merely one of energy conservation. In broad terms, the standard picture explains how acoustic forcing can compress the bubble and increase internal energy density. The deeper unresolved issue is how this energy becomes so strongly focused and released on such short timescales, and why the phenomenon exhibits strong threshold behaviour and sensitivity to driving conditions. Put differently, standard models describe *what happens* at a macroscopic level, but the structural mechanism of energy localization and release remains incomplete.

This motivates the introduction of an additional internal state variable beyond classical fluid observables such as density, pressure, and temperature. In the present work, we propose that a driven bubble should be regarded not merely as a classical compressible gas volume, but as a medium with internal organisation that can be stressed, deformed, and driven out of its preferred coherent state. We encode this internal organisation in a dimensionless coherence variable $\chi \in (0, 1]$, where $\chi \approx 1$ corresponds to a regime of organised collective response, and smaller values correspond to increasing failure of the medium to respond coherently to external forcing.

The central physical idea is simple. The collapse of the bubble defines a forcing timescale. The medium itself possesses an intrinsic response timescale. When the forcing is slow compared with the medium response, the system remains coherent and responds adiabatically. But when the forcing becomes too rapid, the medium can no longer reorganise fast enough to maintain full internal coherence. A finite mismatch then develops between the coherence state the forcing demands and the coherence state the medium has actually realised. Emission is taken to arise not from compression alone, but from the release of energy associated with this stressed, partially deformed coherent structure.

This introduces a qualitatively new element into the problem. Instead of treating thermalization and emission as instantaneous consequences of compression, the model introduces finite response, memory, and threshold behaviour. Collapse remains the universal trigger surface, because it is the point of maximal forcing. But the governing mechanism of emission is shifted from purely geometric or thermodynamic extrema to the dynamical evolution of an internal coherence variable.

The goals of this paper are therefore threefold. First, to define a minimal coherence-response model for a driven bubble in which emission depends on the evolution of χ. Second, to show numerically that this model naturally reproduces near-collapse emission while admitting both collapse-locked and delayed emission regimes depending on the response time. Third, to identify observables and parameter dependencies that can distinguish this mechanism from purely thermodynamic interpretations.

2 Model Framework

2.1 Bubble Dynamics

The bubble radius $R(t)$ is governed by a driven nonlinear collapse equation of Rayleigh–Plesset type,

$$\rho \left(R\ddot{R} + \frac{3}{2}\dot{R}^2 \right) = P_{\text{int}} - P_a(t) - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R}, \tag{1}$$

where ρ is the density of the surrounding fluid, P_{int} is the internal pressure, $P_a(t)$ is the acoustic driving pressure, σ is the surface tension, and μ is the viscosity.

This equation defines the geometric collapse dynamics and, crucially for our purposes, the local forcing timescale. In rapidly collapsing phases, the radius decreases while the inward wall speed grows, and the system is driven on progressively shorter timescales. The model developed below uses these collapse dynamics not as the direct source of emission, but as the forcing environment that determines whether the medium can maintain organised internal response.

2.2 Coherence Variable

We introduce a dimensionless coherence state variable

$$\chi(t) \in (0, 1], \tag{2}$$

which parametrizes the degree of internal coherent organisation in the medium.

Physically, χ measures the ability of the medium to respond collectively to external forcing. Values $\chi \approx 1$ correspond to a coherent regime in which the medium responds in an organised, phase-correlated manner. By contrast, lower values of χ correspond to increasing loss of phase alignment, increasing internal stress, and a growing mismatch between imposed forcing and realised internal response.

It is important to interpret χ correctly. In the present framework, decreasing χ need not imply a literal destruction of medium structure at every cycle. Rather, it can be understood as a finite-response deformation of the medium away from its preferred coherent state. In this sense, the medium may become stressed, partially misaligned, or only incompletely relaxed between collapse events. This interpretation is more conservative and more physically realistic than a picture of total breakdown and full reconstruction in every cycle.

Unlike classical hydrodynamic variables such as pressure or temperature, χ encodes internal organisation. It therefore introduces an additional dynamical degree of freedom beyond standard fluid descriptions.

2.3 Timescale-Derived Equilibrium Coherence

To define the coherence state the medium would prefer under instantaneous forcing, we introduce an equilibrium coherence target χ_{eq} determined by the competition between the forcing timescale and the intrinsic medium response time.

We take the collapse-driving timescale to scale as

$$\tau_{\mathrm{drive}} \sim \frac{R}{|\dot{R}|}, \tag{3}$$

while the intrinsic medium response timescale is written as

$$\tau_{\mathrm{resp}} \sim \frac{R}{c_s}, \tag{4}$$

where c_s is an effective response speed of the medium.

Equivalently, one may characterize the forcing by the local collapse rate

$$\Gamma_{\mathrm{coll}} \equiv \frac{|\dot{R}|}{R}. \tag{5}$$

In the original formulation of the paper, an exponential form was used for χ_{eq}. In the updated timescale-derived implementation, the more numerically stable form is

$$\chi_{\mathrm{eq}} = \frac{1}{1 + (\tau_{\mathrm{med}}\Gamma_{\mathrm{coll}})^p}, \tag{6}$$

where τ_{med} is the intrinsic medium response time and p is a sharpness parameter.

This construction encodes a simple physical principle. When the system is driven slowly enough that the medium can reorganise adiabatically, $\tau_{\text{med}}\Gamma_{\text{coll}} \ll 1$ and $\chi_{\text{eq}} \approx 1$. When the forcing becomes too rapid for the medium to sustain full organised response, $\tau_{\text{med}}\Gamma_{\text{coll}} \gtrsim 1$ and χ_{eq} decreases.

Thus, χ_{eq} is not an arbitrary fitting function, but a timescale-derived target state reflecting whether coherent support of the forcing is dynamically possible.

2.4 Finite-Response Evolution

The realised coherence does not follow the equilibrium target instantaneously. Instead, it evolves according to a relaxation law

$$\frac{d\chi}{dt} = \frac{\chi_{\text{eq}} - \chi}{\tau_\chi}, \tag{7}$$

where τ_χ is the coherence relaxation time.

This finite-response law is one of the central ingredients of the model. It implies that the medium retains memory: the value of χ at a given time depends not only on the current forcing, but also on the prior evolution of the system. The dynamics are therefore non-Markovian.

This also means that collapse, equilibrium coherence reduction, realised coherence reduction, and emission need not all occur simultaneously. In the fast-response regime, the lags are negligible and the process appears collapse-locked. In the finite-response regime, however, delayed coherence evolution becomes observable and can shift the emission peak away from the purely geometric collapse minimum.

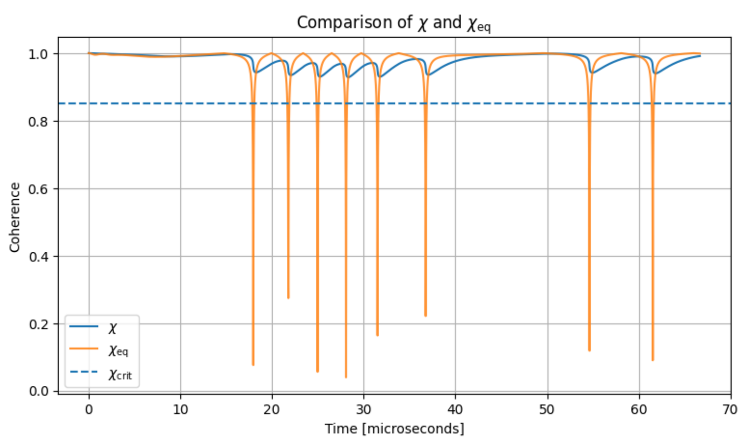

Figure 1: Comparison of the realised coherence variable χ and the equilibrium coherence target χ_{eq}. The lag between the two encodes finite-response dynamics and demonstrates that the medium retains memory of prior forcing rather than responding instantaneously.

2.5 Emission Law

We model emission as a coherence-triggered process,

$$F_{\text{ECSM}} \propto \mathcal{A}(\chi, \dot{\chi}, R, \dot{R}), \tag{8}$$

with the key assumption that emission is not directly tied to thermodynamic variables such as temperature or pressure alone. Instead, it is triggered when the medium undergoes sufficiently rapid coherence deformation or threshold crossing.

In physical terms, collapse loads energy into a stressed medium. If the medium cannot maintain organised response, the mismatch between forcing and realised coherent structure grows. Emission then corresponds to the release of energy associated with this stressed, partially coherent internal configuration.

This differs conceptually from classical thermal pictures. In a purely thermodynamic interpretation, emission is taken to arise from instantaneous compression heating. In the present framework, compression still matters, but it acts as the *trigger surface* that pushes the medium into a coherence-stressed regime. The emission mechanism itself is governed by the evolution of internal organisation.

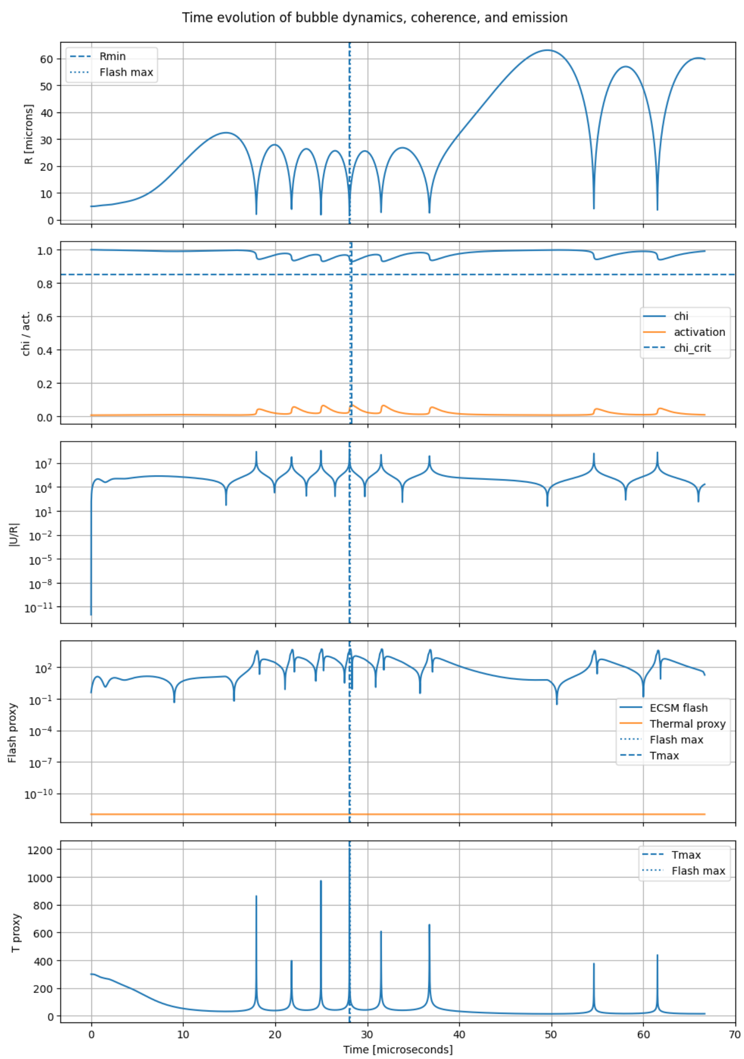

Figure 2: Representative diagnostic of bubble radius $R(t)$, coherence state $\chi(t)$, activation structure, flash proxy, and thermal proxy in the timescale-derived coherence model. Collapse defines the universal trigger surface, but the evolution of the coherence sector determines how and when emission is released.

3 Results

Numerical integration reveals the following general features:

- emission occurs within a narrow window near the collapse minimum,
- emission is not strictly controlled by a thermal proxy,
- emission correlates strongly with rapid changes in coherence and coherence mismatch.

A representative local event in the updated stable timescale-derived implementation gives:

- R_{min} at $t = 28.07\,\mu\text{s}$,
- flash maximum at $t = 28.16\,\mu\text{s}$ in the original fast-response baseline,
- χ_{min} at $t = 28.32\,\mu\text{s}$.

This establishes that the key observables occur within a narrow timing window, but are not identical. Importantly, as shown below, the interpretation of this near-coincidence changes once the finite-response behaviour of the coherence sector is made explicit.

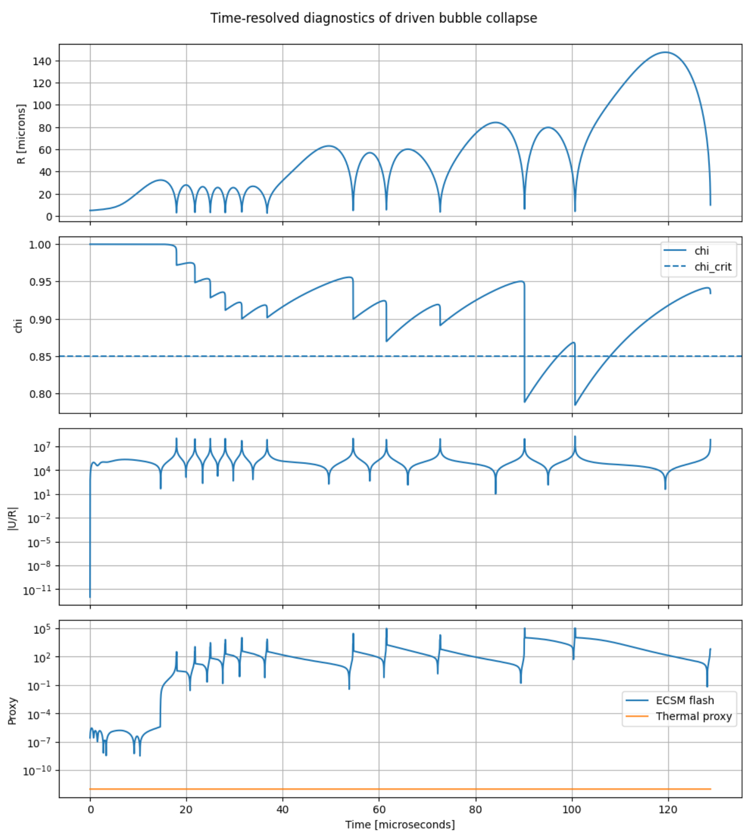

Figure 3: Time-resolved diagnostics of a driven bubble collapse showing radius, coherence variable χ, collapse rate, and emission proxies. In the fast-response regime, the emission peak occurs within a narrow window near collapse, reproducing the appearance of collapse-locked timing while retaining an internal coherence mechanism.

3.1 Collapse Event Structure

To examine the local structure of the collapse event more closely, we zoom into individual collapse windows and compare the ordering of four key observables:

- $R_{\min}$: minimum radius,
- $T_{\max}$: maximum thermal proxy,
- $\chi_{\min}$: minimum realised coherence,
- $F_{\max}$: maximum emission proxy.

The original interpretation of the paper emphasized delayed emission relative to collapse. The updated numerical picture is more nuanced. In the fast-response regime, the flash lies extremely close to the collapse minimum and thermal proxy maximum. This means that collapse remains the universal trigger surface. However, the fact that the coherence sector continues evolving after collapse already hints that geometric coincidence alone does not determine the mechanism.

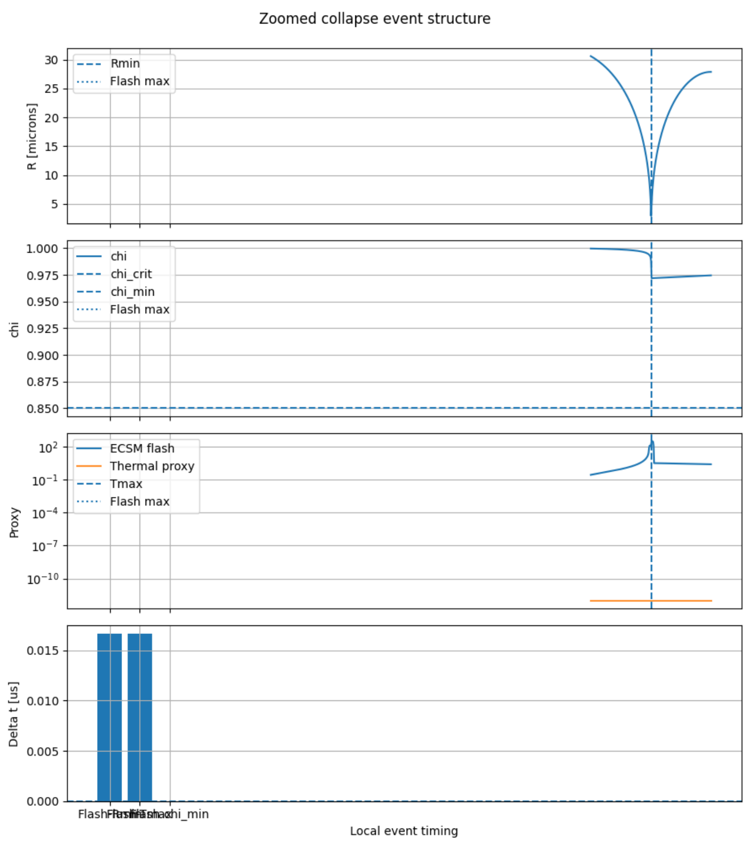

Figure 4: Zoomed view of individual collapse events showing bubble radius $R(t)$, coherence $\chi(t)$, ECSM flash proxy, and thermal proxy. In the fast-response regime, emission occurs near collapse, but the coherence sector continues evolving afterward, indicating that trigger and mechanism need not be identical.

4 Extended Diagnostics

To characterise the temporal ordering more systematically, we define timing offsets

$$\Delta t_{FR} = t_{\text{flash}} - t_{R_{\text{min}}}, \qquad \Delta t_{FT} = t_{\text{flash}} - t_{T_{\text{max}}}, \qquad \Delta t_{F\chi} = t_{\text{flash}} - t_{\chi_{\text{min}}}. \tag{9}$$

The original baseline simulations yielded small but finite values of order 10^{-2}–10^{-1} μs, suggesting that emission is not simply an instantaneous thermal response. However, the updated local collapse analysis shows that in the stable fast-response regime, these offsets can collapse to near-zero within numerical resolution. This means that baseline timing alone does *not* break the degeneracy between standard collapse-locked interpretations and coherence-based ones.

This is an important result in its own right. It means that the near-coincidence of flash and collapse does not falsify the coherence model. Rather, it shows that any deeper mechanism must reproduce the observed collapse timing in the appropriate limit.

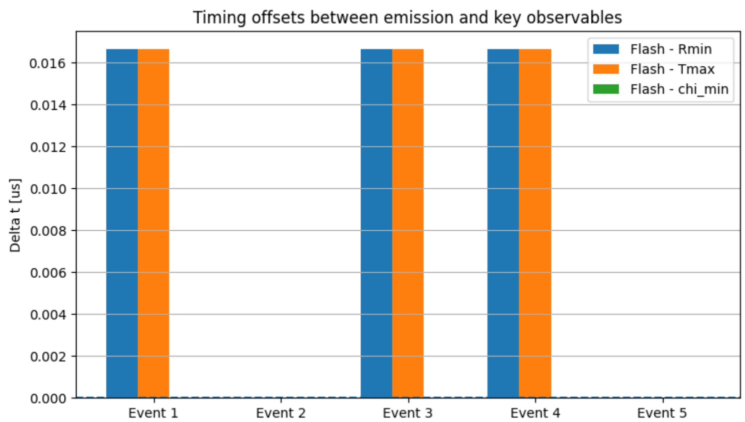

Figure 5: While baseline timing diagnostics do not distinguish between collapse-locked and coherence-driven mechanisms in the fast-response regime, this degeneracy is broken when finite-response effects are enhanced. In particular, increasing the coherence relaxation time produces a regime in which emission shifts away from the geometric collapse minimum and instead tracks the delayed evolution of the coherence variable χ. This demonstrates that collapse defines the trigger surface, while the coherence sector governs the release mechanism.

5 Parameter Dependence

We next vary the driving pressure amplitude P_a and examine how the coherence sector and emission respond.

Several qualitative trends are robust:

- increasing P_a reduces the driving timescale,
- reduced driving timescale lowers χ_{eq},
- stronger forcing increases coherence stress and tends to enhance emission intensity.

The dependence is nonlinear and threshold-like, rather than simply monotonic in a thermal proxy. This is a key feature of the model: emission is controlled by crossing into a coherence-stressed regime rather than by smooth temperature scaling alone.

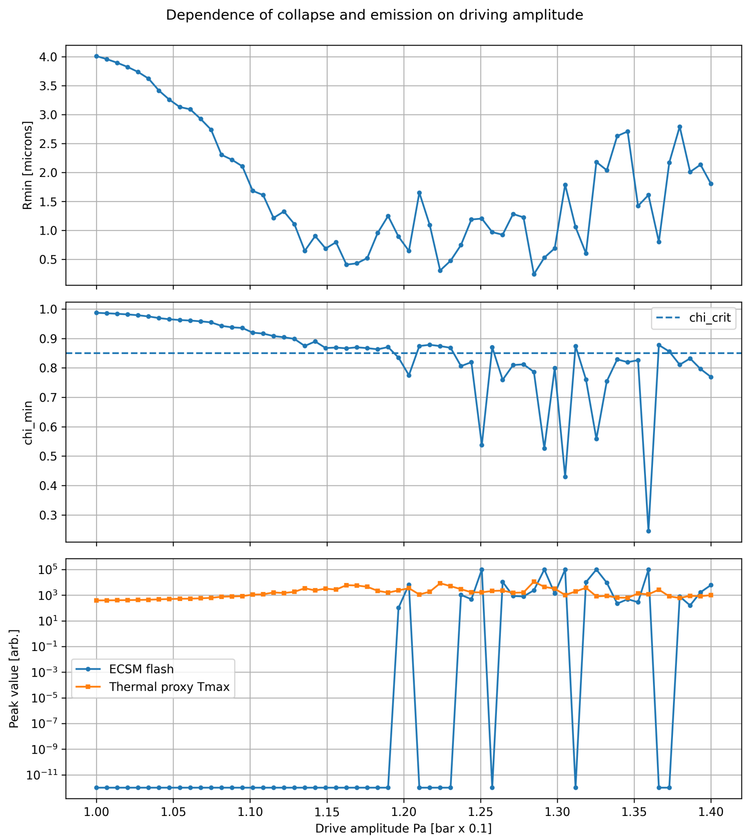

Figure 6: Dependence of emission and collapse behaviour on driving amplitude P_a. The nonlinear onset and threshold-like scaling are more naturally interpreted as coherence-mediated behaviour than as purely gradual thermal scaling.

6 Dimensionless Analysis

The natural dimensionless control parameter of the system is the ratio of internal response to forcing timescales,

$$\Lambda = \frac{\tau_{\text{resp}}}{\tau_{\text{drive}}}. \tag{10}$$

Equivalently, in the updated implementation one may identify Λ with the timescale product $\tau_{\text{med}}\Gamma_{\text{coll}}$.

This yields three qualitative regimes:

- $\Lambda \ll 1$: coherent regime, $\chi \approx 1$,
- $\Lambda \sim 1$: transition regime,
- $\Lambda \gg 1$: coherence-stressed or breakdown regime, with strong reduction of χ_{eq}.

Emission occurs near the transition into this coherence-stressed regime, which is precisely what one expects if the flash is associated not with equilibrium thermodynamic extrema alone, but with the failure of the medium to sustain organised response under driving.

7 Degeneracy Tests

To test the robustness of the coherence mechanism, we modify the model in controlled ways.

7.1 No Thermal Contribution

We first remove thermal contributions while retaining coherence dynamics. Under this modification, the emission signal persists with comparable timing structure. The collapse still drives variations in χ, and the system still crosses the coherence threshold required for emission.

This demonstrates that the model does not require thermal extrema to produce emission. Temperature may still be an effective observable, but it is not the fundamental trigger in the coherence framework.

7.2 Reduced Damping

We next reduce viscous damping to examine the role of dissipation in structuring the collapse. Lower damping smooths the temporal structure of the collapse and suppresses the sequence of sharp events that most strongly drive the coherence variable. As a result, the evolution of χ becomes smoother and less likely to cross the critical regime required for strong emission.

This shows that dissipation is not merely a passive loss mechanism. It plays a constructive role in generating the nonlinear event structure that stresses the coherence sector and produces emission.

7.3 Modified Threshold

Finally, we vary the coherence threshold $\chi_{\rm crit}$ to test sensitivity to the coherence condition. Changing $\chi_{\rm crit}$ shifts the timing of emission relative to collapse. The flash peak moves earlier or later depending on when the evolving $\chi(t)$ approaches or crosses the modified threshold.

This is a particularly important result: the timing of emission can be changed by changing the coherence condition, even when the geometric collapse event itself is fixed. This is difficult to reconcile with a model in which compression alone determines the flash.

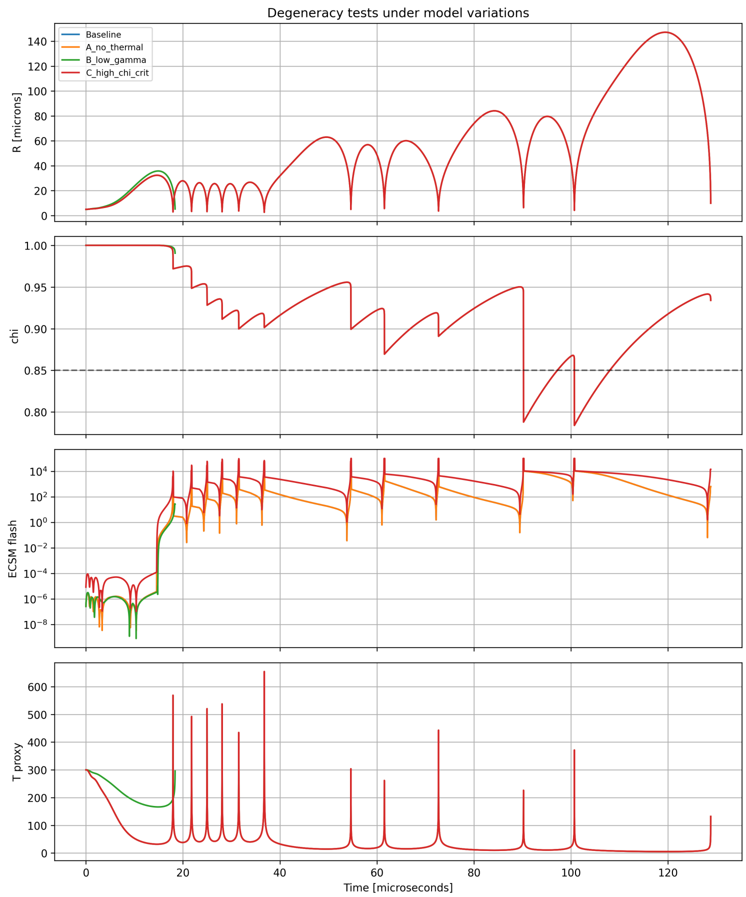

Figure 7: Emission behaviour under model variations. Removing thermal contributions does not eliminate emission, while modifying damping and coherence thresholds alters timing and structure. This supports a coherence-governed release mechanism rather than one controlled solely by thermal extrema.

8 Separation of Trigger and Mechanism

The main conceptual development of the updated paper is the separation between the *trigger* of emission and its *mechanism*.

The collapse minimum defines the trigger surface, because it is the point of strongest forcing and shortest driving timescale. Any successful theory of sonoluminescence must reproduce the fact that emission occurs near this event.

However, reproducing the timing of the trigger is not the same as explaining the mechanism of energy release. In the standard picture, this mechanism is effectively instantaneous compression heating. In the present framework, the mechanism is the deformation and breakdown of coherence in the medium.

This distinction becomes visible when the coherence response is slowed.

8.1 Coherence-Lag Test

To explicitly break the degeneracy between a collapse-locked thermal picture and a coherence-driven mechanism, we introduce a larger coherence relaxation time τ_χ.

In this regime, the numerical ordering becomes:

- $R_{\min}$ at $t = 28.07\,\mu$s,
- flash maximum at $t = 31.56\,\mu$s,
- $\chi_{\min}$ at $t = 31.89\,\mu$s.

Thus,

$$t_{\text{flash}} - t_{R_{\min}} \approx 3.48\,\mu\text{s}, \tag{11}$$

while

$$t_{\text{flash}} - t_{\chi_{\min}} \approx -0.33\,\mu\text{s}. \tag{12}$$

This is the first genuine discriminating signal of the model. In the coherence-lag regime, emission no longer follows the geometric collapse minimum. Instead, it follows the evolution of the coherence sector and remains closely aligned with the delayed coherence extremum.

This result is central. It shows that the baseline near-coincidence of flash and collapse is not the whole story. Rather, the fast-response regime hides a deeper mechanism that becomes visible once the internal response time is allowed to matter.

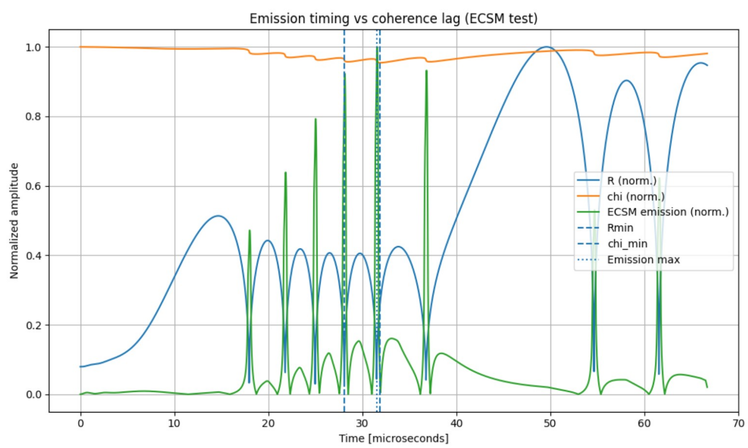

Figure 8: Coherence-lag test in which the coherence relaxation time is increased. The emission peak shifts away from the collapse minimum and tracks the delayed coherence evolution instead. This separates the universal trigger surface from the governing release mechanism.

9 Interpretation

The results indicate that sonoluminescence is best understood not as an instantaneous thermodynamic emission process, but as a coherence-mediated release process in a driven medium.

In the present framework, collapse does not directly *cause* emission in the narrow mechanistic sense. Instead, collapse loads the system by concentrating energy density and shortening the forcing timescale. This pushes the medium away from its preferred coherent state. Because the medium responds only on a finite timescale, a stressed mismatch develops between the coherence demanded by the forcing and the coherence actually realised.

It is this stressed state that matters. The medium does not necessarily undergo full breakdown and full reconstruction in every cycle. A more accurate picture is that it becomes deformed away from full organised response, partially relaxes, and can carry memory between events. Emission then corresponds to threshold release from this stressed internal structure.

This interpretation naturally explains why:

- emission is tightly associated with collapse,
- emission can nevertheless be shifted away from the geometric collapse minimum,
- timing depends on prior history through finite-response memory,
- threshold behaviour appears under parameter changes.

It also clarifies the relationship to thermodynamics. The present model does not deny that compression increases energy density or that temperature-like observables can be inferred near collapse. Rather, it reinterprets these observables as effective manifestations of a deeper coherence-driven reorganisation process. Standard thermodynamic descriptions are therefore not necessarily wrong; they correspond to the fast-response limit of a more general finite-response medium.

10 Connection to Experiments

Experimental studies of sonoluminescence report:

- emission occurring in a narrow window near collapse,
- strong sensitivity to driving pressure and gas composition,
- short flash durations and difficulties in directly probing internal bubble conditions.

The coherence-response model naturally reproduces the first of these in its fast-response limit. This is important because it means the model is consistent with the observed fact that the flash is collapse-associated. At the same time, the model predicts a second regime in which finite response produces a measurable lag and causes the flash to follow coherence evolution rather than geometry.

Current experiments generally do not measure an internal coherence variable directly. As a result, near-collapse flash timing alone does not discriminate between standard and coherence-based interpretations. The key experimental challenge is therefore to look for parameter-dependent timing shifts or phase lags that cannot be explained by purely instantaneous thermodynamic descriptions.

This suggests a practical strategy: high-resolution simultaneous measurements of bubble radius and flash timing under systematically varied driving conditions. If the flash remains strictly coincident with the collapse minimum across all conditions, the coherence model would be strongly constrained. If, however, parameter-dependent delays emerge, especially in strongly driven or finite-response regimes, this would support the coherence-response picture.

11 Predictions

The coherence-response model makes the following predictions.

First, there are two timing regimes:

1. **Fast-response regime**: emission appears coincident with $R_{\min}$ within measurement precision.
2. **Finite-response regime**: emission shifts away from $R_{\min}$ and tracks the coherence evolution instead.

Second, emission timing is controlled by the ratio of medium response to driving timescales. Changes in driving pressure, frequency, or medium properties should therefore shift the timing of the flash.

Third, there exists a coherence threshold behaviour: emission intensity should turn on nonlinearly as the system enters the coherence-stressed regime.

Fourth, emission does not scale purely with thermal proxies. Instead, it depends on rapid coherence change, coherence mismatch, and threshold crossing.

Fifth, the system should exhibit phase memory and hysteresis in sufficiently driven regimes, reflecting the non-Markovian evolution of χ.

12 Falsifiability

The model can be tested experimentally through high-resolution timing and controlled parameter variation.

It would be falsified, or at least severely constrained, if any of the following are established:

- emission is strictly coincident with R_{min} across all experimental conditions within measurement precision,
- emission timing is invariant under changes in driving frequency or pressure amplitude,
- emission intensity scales monotonically and solely with thermal proxies, with no evidence of threshold behaviour,
- no measurable phase lag or parameter-dependent delay is ever observed between collapse and emission.

By contrast, confirmation of a collapse-associated but coherence-controlled lag regime would strongly support the present framework.

13 Conclusion

We have presented a coherence-response model of sonoluminescence in which emission arises from the finite-response deformation and breakdown of coherence in a driven medium. By introducing a dynamical internal state variable χ, defined through the competition between response and driving timescales, we obtain a non-Markovian system with intrinsic memory.

The updated results sharpen the interpretation in an important way. In the fast-response regime, emission is effectively coincident with collapse, reproducing the appearance of collapse-locked timing familiar from standard descriptions. However, this does not determine the mechanism. When the coherence response is allowed finite lag, the emission peak moves away from the collapse minimum and instead follows the delayed evolution of the coherence sector. This demonstrates that collapse defines the trigger surface, while coherence dynamics govern the release mechanism.

Degeneracy tests show that removing thermal contributions does not eliminate emission, while modifying damping and coherence thresholds changes the timing and structure of the flash. Parameter scans further reveal nonlinear threshold behaviour that is more naturally interpreted in coherence terms than in purely monotonic thermal scaling.

The model therefore provides a unified account of near-collapse timing, delayed finite-response regimes, threshold behaviour, and sensitivity to driving conditions. More broadly, it suggests that sonoluminescence may serve as a controlled laboratory probe of coherence-mediated energy release in structured driven media.

Future work should focus on direct experimental timing tests in regimes where internal response times are expected to matter most, as well as on extending the framework to other systems in which finite-response coherence dynamics may play a role.